

EduGenie: Google Gemini Powered Learning Assistant

5. Project Development Phase

This document records the development activities and implementation responsibilities described in the EduGenie source material.

5.1 Development Structure

The implementation is divided into independent Python modules and a simple HTML/CSS frontend.

FastAPI connects the user interface to the appropriate module through RESTful endpoints.

5.2 Explanation Development

The explanation capability uses LaMini-Flan-T5-783M.

The documented objective is to produce concise, context-aware explanations that break down complex topics into accessible language.

The module is positioned for beginners, school students, and self-learners.

5.3 QnA Development

Gemini 1.5 Pro provides general knowledge and academic question answering.

The module is exposed through the /qa endpoint.

5.4 Quiz Development

The quiz module generates three MCQs from a passage.

Each question contains four options.

The implementation requests structured JSON.

Markdown code blocks are cleaned before parsing using clean_json_block.

Generation or parsing failures are returned as detailed error messages for debugging.

5.5 Summary Development

Gemini is used to summarize long paragraphs.

The goal is concise revision material that retains core information while reducing redundancy.

5.6 Learning Path Development

The get_learning_recommendations function creates a personalized structured path.

The prompt requests beginner-to-advanced concepts organized by difficulty.

Suggested resources may include videos, articles, or books.

The function handles invalid or incomplete API responses with a helpful error message.

5.7 API Development

/qa routes Q&A.;

/explain routes concept explanation.

/quiz routes quiz generation.

/summarize routes summarization.

/learn/recommendations routes learning recommendations.

5.8 Frontend Integration

The web interface provides a task selector, text area, and submit button.

Form submission sends a POST request to FastAPI.

The result appears below the input area in real time.

Conclusion

The development phase converts the design into a modular application where each AI capability can be developed and tested independently while FastAPI provides a consistent integration layer.

Source basis: EduGenie project document supplied with this request and the repository structure shown in the supplied reference image. Where the source did not specify an implementation detail, this documentation preserves the documented scope rather than inventing project facts.